

CHAPTER 02 CABLES

II.1. Introduction

II.2. Characteristics and working mode of a cable

II.3. Cable subjected to concentrated loads

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II.1. Introduction

Cables are important and efficient structural elements that primarily carry loads through tensile forces. Their high tensile strength and relatively low self-weight make them widely used in suspension and cable-stayed bridges, roofs, towers, elevators, cable cars, and lifting equipment.

From a structural point of view, an ideal cable is assumed to be **perfectly flexible and inextensible**. It cannot resist bending, shear, or compression ; therefore, the internal force is always tensile and tangent to the cable at every point. The shape of a cable depends on the applied loads and the support conditions.

Depending on the loading conditions, different cable configurations may be obtained. This chapter considers cables subjected to **concentrated loads, uniformly distributed loads along the horizontal projection, and uniformly distributed loads along the cable itself**. Although these loading conditions produce different cable geometries, their equilibrium can be established using the fundamental principles of statics. The relationships between the applied loads, support reactions, tensile forces, cable slope, and geometry are developed accordingly.

The analysis of cables is essential for determining **support reactions, internal tensile forces, cable profiles, slopes, and displacements**. The chapter also establishes relationships between the horizontal component of cable tension, the internal forces, the cable geometry, and its total length. Finally, the influence of cable elasticity is considered when the assumption of inextensibility is relaxed.

These concepts provide a fundamental basis for understanding and analyzing cable-supported structures in civil and structural engineering.

II.1.1. Pedagogical Objectives

Upon completion of this chapter, students will be able to :

1. **Define** a structural cable and explain its fundamental mechanical role in a structural system.
2. **Identify** the main applications of cables in civil engineering, including suspension and cable-stayed bridges, suspended roofs, guyed towers, and overhead transmission lines.
3. **Explain** the mechanical behavior of an ideal cable and understand why it carries only axial tension with **zero bending moment** ($M=0$).
4. **Establish and apply** the fundamental equations governing cable equilibrium and geometry under different loading conditions.
5. **Analyze** cables subjected to concentrated vertical loads using equilibrium equations at discrete points.
6. **Determine** the cable profile, sag, horizontal tension (H), and maximum tension (T_{max}) for different types of distributed loading, including the parabolic and catenary configurations.
7. **Determine** support reactions and analyze different cable support configurations, such as pulleys, saddles, and roller supports.
8. **Apply** cable analysis principles to the preliminary sizing and structural assessment of practical cable-supported systems.

II.2. Characteristics and Working Behavior of a Cable

Cables have several specific characteristics that distinguish them from conventional structural members.

II.2.1 Characteristics of cables

II.2.1.1 High Strength-to-weight ratio

One of the main advantages of cables is their high tensile strength compared with their relatively low weight. This characteristic makes them particularly efficient for structures subjected mainly to tensile forces and for applications involving long spans.

II.2.1.2 High tensile strength

Cables are manufactured from high-strength steel wires. Depending on the steel grade and cable construction, high strength classes can be obtained.

The cable catalog specifies, for example, standard steel grades of approximately:

- 1770 N/mm²;
- 1960 N/mm²;
- 2160 N/mm².

These values correspond to the nominal tensile strength classes of the steel used for the cable wires.

II.2.1.3 Flexibility

A cable has a very small diameter compared with its length. Consequently, it is highly flexible and can adapt to curved configurations under loading.

The flexibility of a cable depends on several factors, including:

- The number of strands;
- The number and diameter of wires in each strand;
- The type of core;
- The arrangement of the wires;
- The lay direction ;
- The cable construction.

Different cable constructions are therefore selected according to the required degree of flexibility and the type of service.

II.2.1.4 Ductility and elongation

Steel cables are manufactured through processes such as wire drawing, which influence their mechanical behavior. Their elongation characteristics therefore differ from those of conventional structural steel members.

Cable selection must take into account the expected elongation under service loading, particularly in applications where deformation can significantly influence the structural response.

II.2.1.5 Resistance to wear and fatigue

Cables may be subjected to repeated bending, vibration, shocks, and changes in curvature during service. These effects can lead to wire wear and fatigue.

The required properties depend directly on the application and operating conditions.

II.2.2 Construction of a steel wire rope (For more details, please refer to the PDF link. : https://drive.google.com/file/d/1eQQINRZF6_IiOx3F-8NP_myZgcpr6Dph/view?usp=sharing)

A steel wire rope is composed of three main components as shown in Fig 1 :

1. **a central core;**
2. **several strands arranged around the core;**
3. **steel wires forming each strand.**

A common construction consists of a central core surrounded by **six or eight outer strands**. The rope construction depends on the number of strands, number and arrangement of wires, core type, direction of lay, and steel grade.

Key terminology

French	English	Meaning
Fil	Wire	Small steel element
Toron	Strand	Group of several wires
Âme	Core	Central supporting element
Câble	Wire rope	Complete assembly

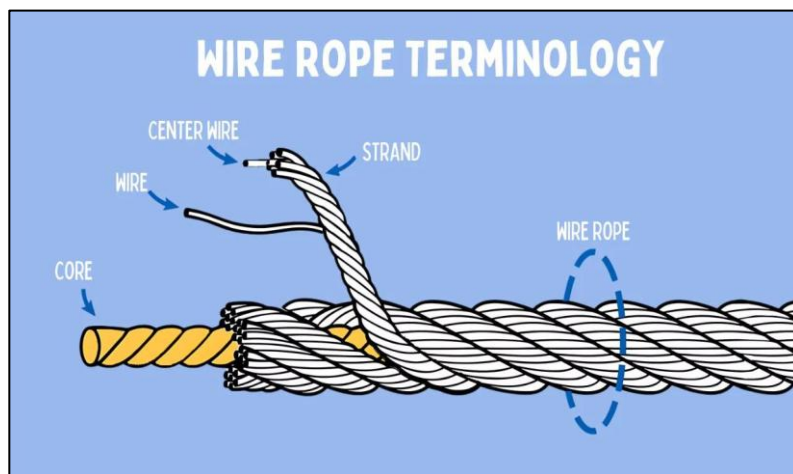
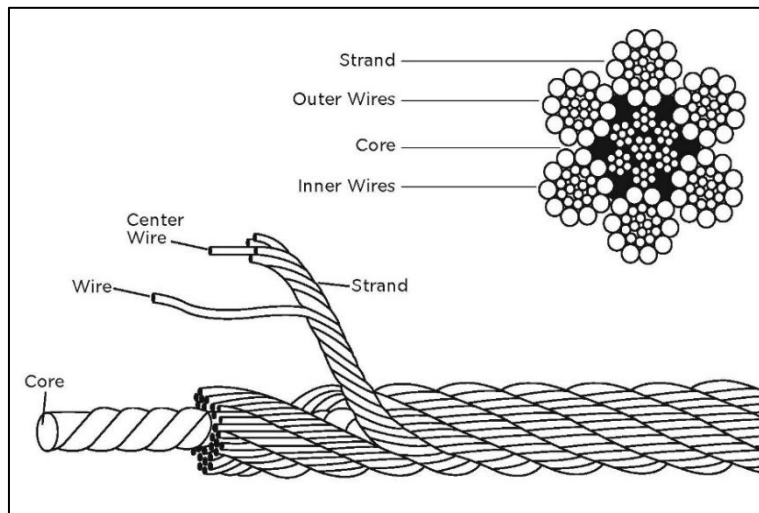


Fig 1. Wire rope terminology.

II.2.2.1 Different types of wire ropes: construction and applications

a) Galvanized Aviation-Type Wire Ropes — 6×19 and 7×19

• Figure A — Construction



• Figure B — Main Applications



Fig.2 — Câbles galvanisés type aviation : construction 6×19 / 7×19 et exemples d'utilisation.

b) Classe 6×19 — EN 12385-4

Figure A — Construction 6×19

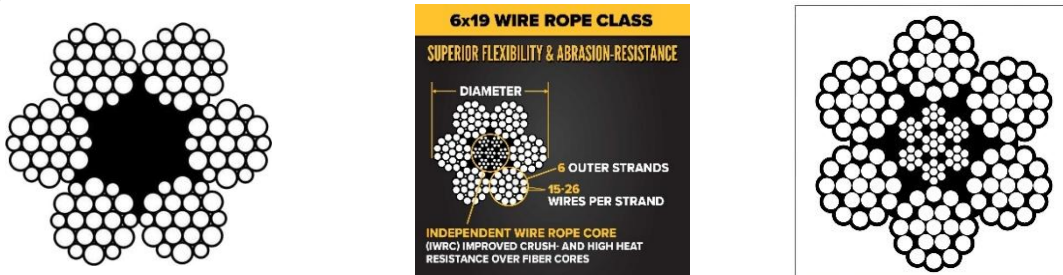


Figure B — Applications: Cranes, Winches, and Logging



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Fig.3 — 6×19 Wire Rope : Construction and Lifting/Winching Applications.

c) Classe 6×36 — EN 12385-4

Figure A — Construction 6×36

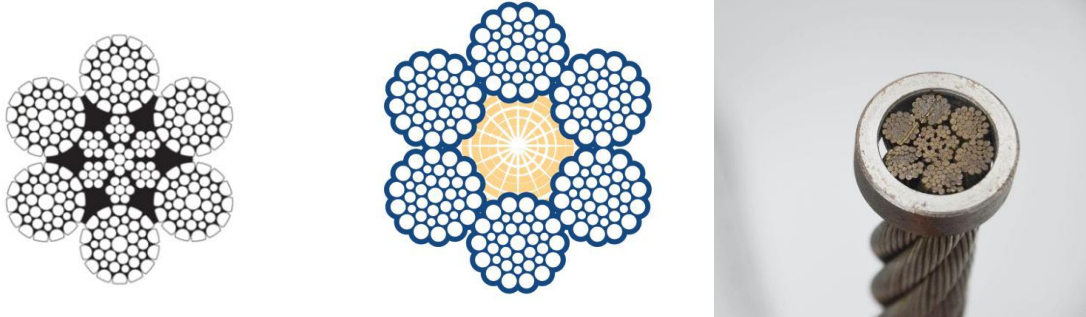


Figure B — Applications: Slings and Lifting

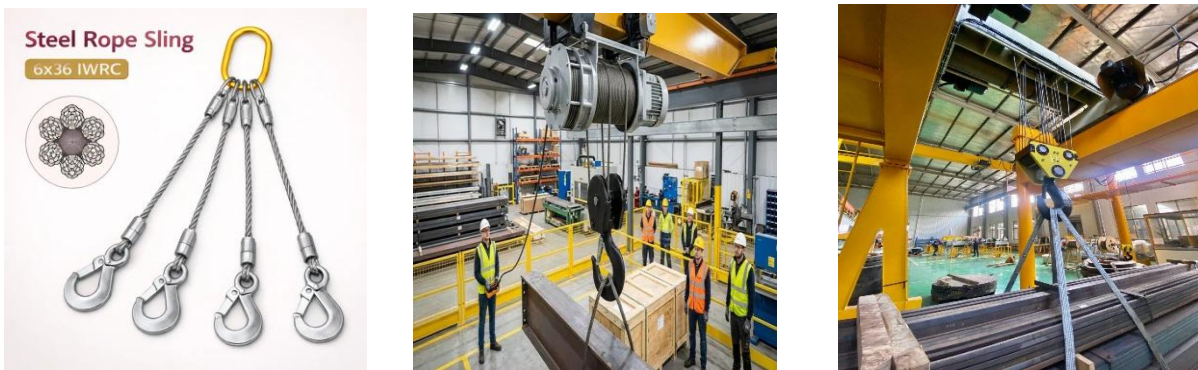


Fig.4 — 6×36 Wire Rope: Flexible Construction and Lifting Applications.

d) Compacted 8×19 and 8×36 Wire Ropes

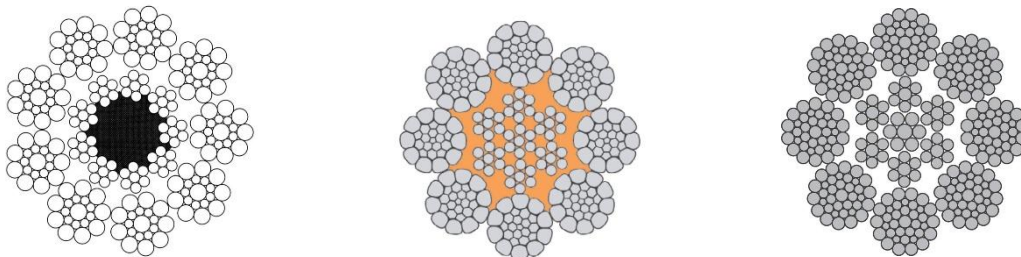


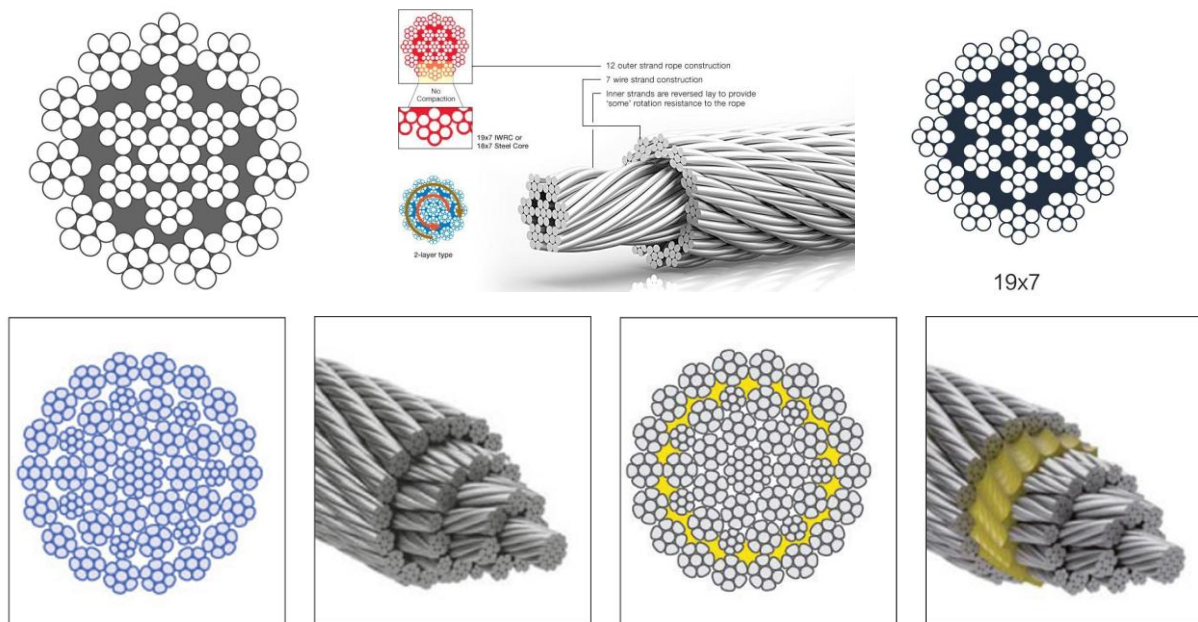
Figure B — Applications on Drums / Cranes



Fig.5 — Compacted Wire Rope: Modified Strand Geometry and Applications in Lifting Equipment.

e) Rotation-Resistant Wire Ropes — 17×7, 18×7, 19×7

Figure A — Construction



f) Locked-Coil Wire Ropes

Applications:

- Suspension bridges
- Stay cables
- Dynamic applications may be considered under certain conditions

Construction:

- One or more layers of **Z-shaped wires**
- Several configurations:
 - **Single layer of Z-shaped wires**
 - **Two layers of Z-shaped wires**
 - **Multi-layer Z-shaped wires**

Figure A — Construction : Single-, double-, and multi-layer Z-shaped wire configurations

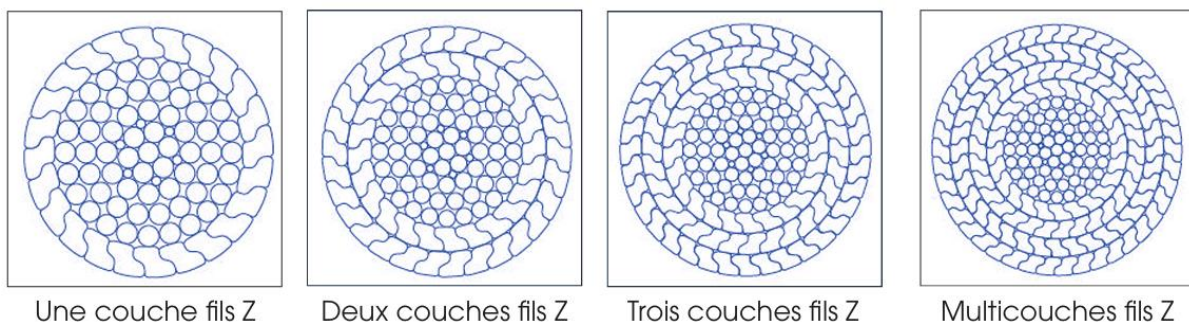


Figure B — Applications : Suspension bridges and stay-cable systems



Fig.5 — Applications : Suspension bridges and stay-cable systems

g) SPECIAL-PURPOSE MONO-STRAND CABLES

Strand cables consist of a single large-diameter strand.

The configurations indicated in the source are :

- 61 wires
- 91 wires
- 127 wires
- 169 wires

Figure A — Construction : Cross-sections of strand cables: 61, 91, 127, and 169 wires.

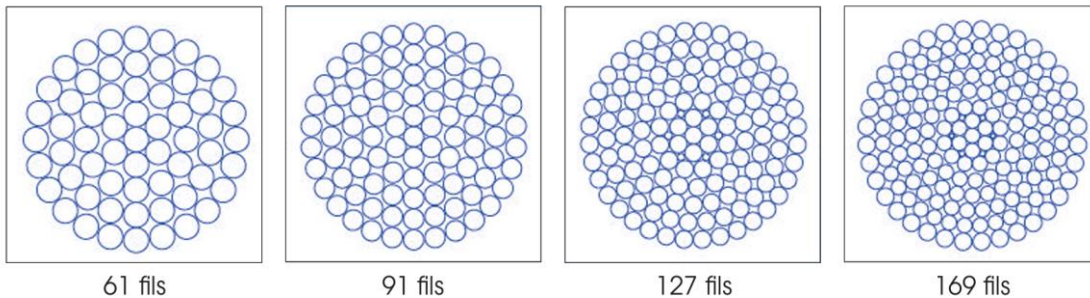


Figure B — Applications : Suspension bridges



Fig.6 — Applications : Suspension bridges

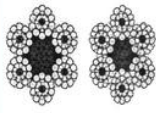
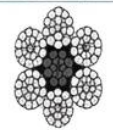
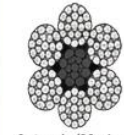
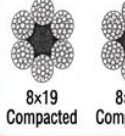
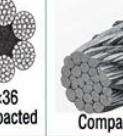
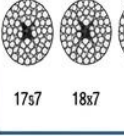
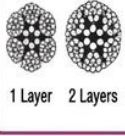
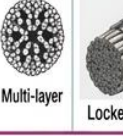
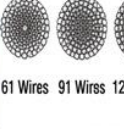
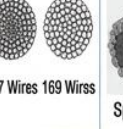
Different Types of Wire Ropes – Construction and Uses				
1. Aviation Type Wire Rope				
Construction / Class	6x19 / 7x19	Primary Application		
		 Small diameter rope applications (controls, mechanisms)		
6 strands (1+6+12)	Galvanized Rope View			
2. 6x19 Class Wire Rope				
Construction / Class	6x19	Primary Application		
		 Logging, skidding, drilling, winches		
6 strands (1+6+12)	6x19 Rope View			
3. 6x36 Class Wire Rope				
Construction / Class	6x36 WSL	Primary Application		
		 Slings, earthmoving, winches, overhead cranes, logging		
6 strands (36 wires)	6x36 Rope View			
4. 8x19 Class Wire Rope				
Construction / Class	8x19 (metallic core)	Primary Application		
		 Cranes, main hoisting, overhead cranes, gantry cranes		
8 strands (1+6+12)	8x19 Rope View			
5. 8x36 Class Wire Rope				
Construction / Class	8x36	Utlimary Application		
		 Mobile cranes, harbor cranes, overhead cranes		
8 strands (36 wires)	8x36 Rope View			
6. Compacted Wire Rope				
Construction / Class	8x19 / 8x36 Compacted	Primary Application		
		 Hoisting and winding equipment		
8x19 Compacted	8x36 Compacted			
7. Non-rotating Wire Rope				
Construction / Class	17x7 / 18x7 / 19x7	Primary Application		
		 Non-rotating Rope View		
17s7	18x7	 Hoists, tower cranes, overhead cranes, winches		
19x7				
8. Locked Coil Rope				
Construction / Class	1 or more layers of Z-shape wires	Primary Application		
		 Locked Coil Rope View		
1 Layer	2 Layers	 Suspension bridges, stay cables		
Multi-layer				
9. Spiral Strand (Single Strand) Rope				
Construction / Class	61 / 91 / 127 / 169 wires	Primary Application		
		 Spiral Strand View		
61 Wires	91 Wires	 Stay cables, suspension bridges		
127 Wires	169 Wires			
10. Stainless Steel Wire Rope				
Construction / Class	1x19 (1+6+12)	Primary Application		
		 Industrial and marine applications (corrosive environment)		
1x19 (1+6+12)	Stainless Steel Rope View			

Fig.7— Different Types of Wire Ropes – Construction and Uses

II.2.3 Cables in Structural Engineering : Theory and Applications

II.2.3.1 Suspension Roofs

A **suspension roof** is a roof in which cables constitute the principal load-carrying elements.

Typical applications include:

- stadiums;
- sports halls;
- exhibition centers;
- railway stations;
- large halls;

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Figure 8— Cable-net suspension roof of Allegiant Stadium, Las Vegas.



Figure 9 — Cable-net roof of SoFi Stadium, Inglewood, California.



Figure 10 — King Fahd International Stadium, Riyadh



Figure 11— Railway station with tensile/cable-supported roof

II.2.3.2 Suspension Bridges :

A **suspension bridge** is a bridge in which the main load-carrying elements are **suspension cables**. The cables are anchored at both ends and pass over towers. Vertical **suspenders (hangers)** connect the main cables to the bridge deck and transfer the loads from the deck to the cables

a) Main structural components

1. **Main cables** — carry the principal tensile forces.
2. **Towers** — support the main cables and transfer vertical and horizontal forces to the foundations.
3. **Hangers / suspenders** — connect the bridge deck to the main cables.
4. **Bridge deck** — carries vehicles, pedestrians, and other traffic loads.
5. **Anchorage** — resist the horizontal component of the cable tension at the ends.
6. **Stiffening girder or truss** — provides rigidity to the deck and distributes traffic loads.

b) Structural principle

The load path can be summarized as:

Traffic load → Deck → Hangers → Main cables → Towers + Anchorages → Foundations

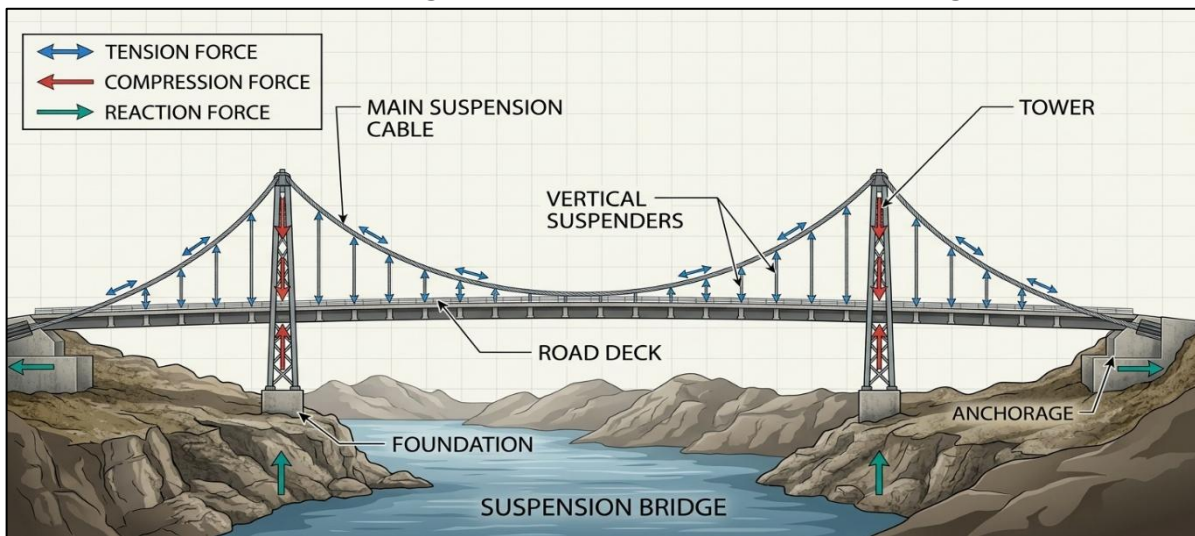


Figure 12- Load Path in a Suspension Bridge (From Traffic to Foundations)

c) Real Applications



Figure 13 — Golden Gate Bridge, San Francisco, USA.



Figure 14 — Akashi Kaikyo Bridge, Japan.

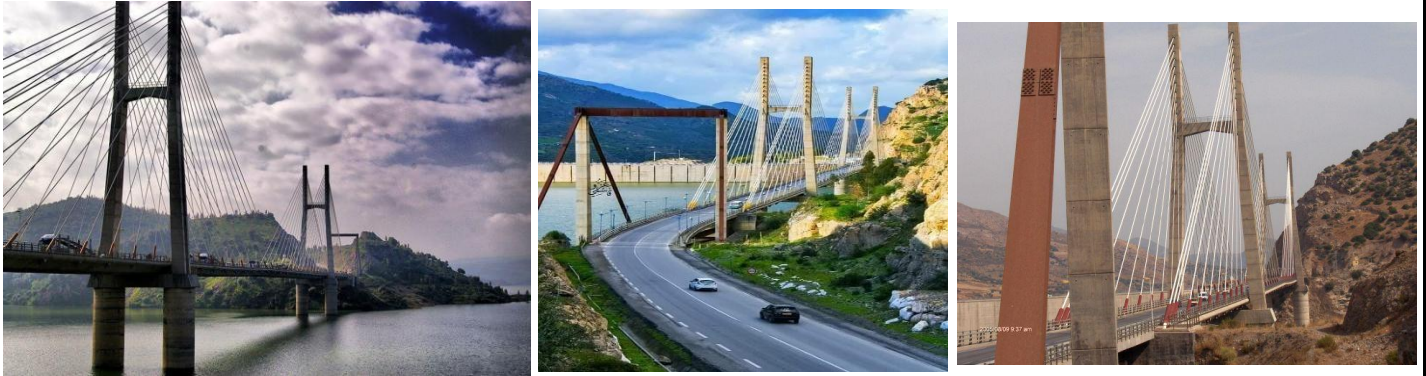


Figure 15 — Sidi Merouane Cable-Stayed Bridge —Mila- Algeria



Figure 16— Main cables and suspenders of Sidi M'Cid Bridge.

II.2.3.3 Guy Wire / Guy Cable : A guy wire is a tensioned cable used to support, stabilize, and brace a structure. It works primarily in tension.



Figure 17- Guyed antenna mast: guy wires stabilize the mast against lateral loads

II.2.3.4 Cable Car / Aerial Cableway : An aerial cableway is a transportation system in which cables support and move cabins or vehicles between stations. The cables carry the weight of the cabins and passengers and transmit the loads to the supporting towers and anchorages.



Figure 18- An aerial cableway

II.2.4 Types of Cable Anchorages

II.2.4.1 - Conical Socket Anchorage

A **conical socket anchorage** consists of a tapered metal socket in which the cable wires are secured. It transfers the cable's tensile force to the supporting structure.



Figure 19 — Conical socket anchorage

II.2.4.2 - Wedge Anchorage :

A wedge anchorage uses wedges to grip and lock the cable. The tensile force is transferred from the cable to the anchorage body through the wedges.



Figure 20— Wedge-type cable anchorage

II.2.4.3 - Clamped Anchorage

A **clamped anchorage** uses mechanical clamps, bolts, or clamping plates to grip the cable and transfer its tensile force to the supporting structure.



Figure 21— Clamped cable anchorage

II.2.4.4 - Ground Anchorage

A **ground anchorage** transfers the tensile force of a cable into the ground through an anchorage block, foundation, or ground anchor.



Figure 22— Ground anchorage for a guy cable

II.2.4.5 - Threaded Anchorage

A **threaded anchorage** uses a threaded connection to secure the cable or its terminal component and transfer the tensile force to the supporting structure.

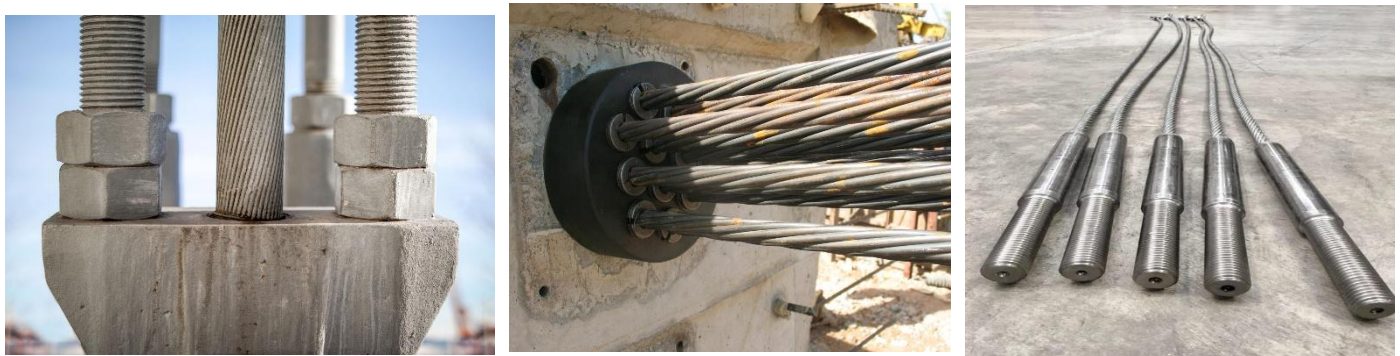


Figure 23 — Threaded cable anchorage

II.2.4.6 - Socket Anchorage

A **socket anchorage** is a cable termination system in which the end of a cable is inserted into a metal socket and securely fixed, usually by a socketing material such as zinc or resin.



Figure 24— Socket anchorage for a structural cable

II.2.5 Basic Assumptions for cable analysis

Cables are **flexible structural elements** mainly used to carry and transmit loads through **tension**. They are widely used in **suspension bridges, cable-stayed bridges, suspension roofs, guyed towers, cableways**, and other tension structures.

For the theoretical analysis of cables, an idealized mechanical model is adopted. This model is based on several simplifying assumptions that allow the equilibrium configuration and internal forces of the cable to be determined.

II.2.5.1 Perfectly Flexible Cable

An ideal cable is assumed to be **perfectly flexible**, meaning that its bending stiffness is negligible. Consequently, the cable can change its curvature freely under the applied loads without developing a bending moment.

Therefore, for a perfectly flexible cable:

- The cable cannot resist **bending moments** ;
- The cable cannot resist **shear forces** ;
- The cable carries only an **axial tensile force** ;
- The tensile force is always **tangent to the cable** at each point.

For an ideal cable, the bending moment and shear force are therefore zero:

$$M = 0$$

$$V = 0$$

The only internal force is the axial tensile force :

$$N = T > 0$$

where M is the bending moment, V is the shear force, and $N = T$ is the tensile axial force.

 **Key idea:** An ideal cable is a tension-only structural element.

II.2.5.2 Inextensible Cable

The cable is assumed to be **inextensible**, meaning that its elastic deformation is neglected. Consequently, the total length of the cable remains constant before and after loading.

If L_0 is the initial length and L is the length after loading :

Or :

$$\Delta L = 0$$

If the cable is represented by the curve

$$y = y(x),$$

The differential length of the cable is:

$$ds = \sqrt{dx^2 + dy^2}$$

Since

$$dy = \frac{dy}{dx} dx,$$

we obtain:

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Therefore, the total length of the cable between supports A and B is:

$$L = \int_{x_A}^{x_B} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

For an **inextensible cable**:

$$\int_{x_A}^{x_B} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = L_0$$

Thus, the cable-length compatibility condition is:

$$L - L_0 = 0$$

Or equivalently:

$$\Delta L = 0$$

 **Key idea:** The cable may change its shape, but its total length remains constant.

II.2.5.3 Neglect of the Cable Self-Weight

In some problems, the **self-weight of the cable** is neglected when it is small compared with the applied loads.

The distributed load due to the cable's own weight is then considered negligible:

$$q_c \cong 0$$

where ($q_c \cong 0$) represents the distributed load intensity due to the self-weight of the cable.

Under this assumption, the analysis considers only the **externally applied loads**.

✚ Important Remark

The self-weight cannot always be neglected. In **long-span cable structures**, such as suspension bridges, the weight of the cable itself can represent a significant part of the total loading and can strongly influence the cable profile and tensile forces.

✚ **Key idea:** The self-weight may be neglected only when its effect is sufficiently small compared with the applied loads.

II.2.6 Types of loads applied to the cable

The loads acting on a cable can generally be divided into two main categories:

II.2.6.1 Concentrated Loads (P_i)

II.2.6.2 Distributed Loads :

- Load per unit horizontal projection $q(x)$
- Load per unit cable Length $q(s)$

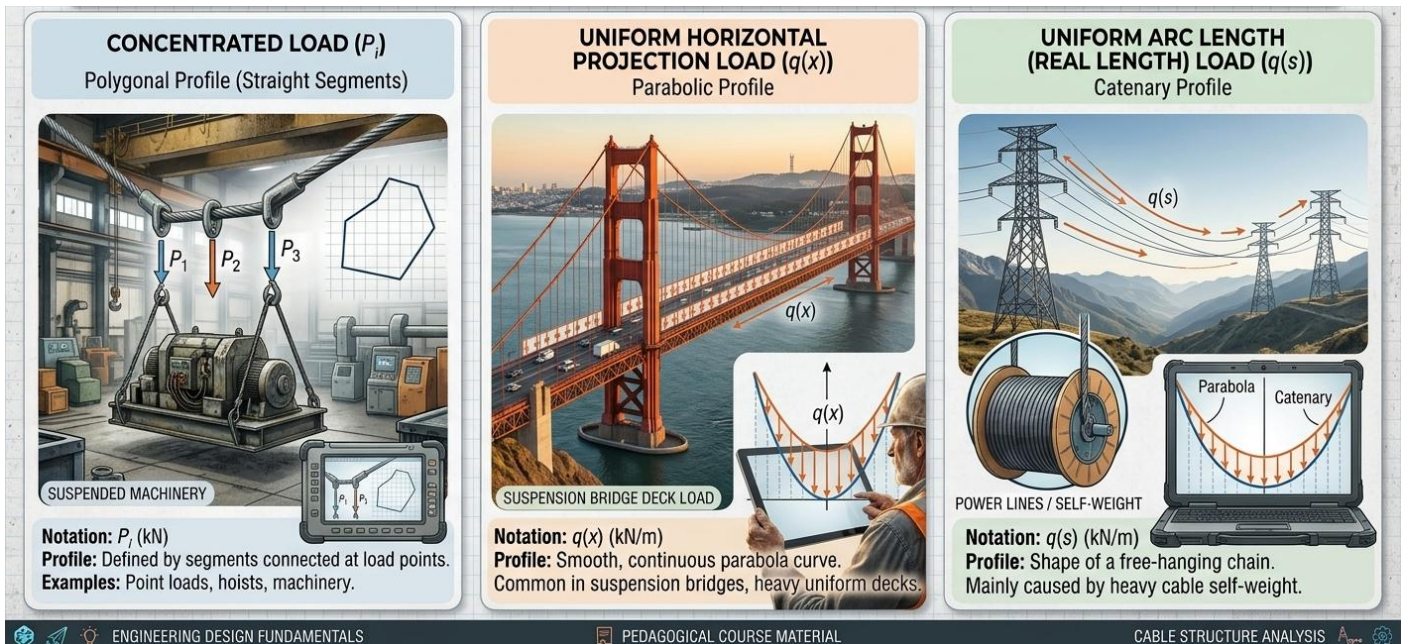


Figure 25— Types of loads applied to the cable

II.3. Cable subjected to concentrated loads

A concentrated load acts at a specific point of the cable. Several concentrated loads can be represented by :

$$P_1, P_2, P_3, \dots \dots \dots$$

where each (P_i) represents a concentrated force.

When a cable is subjected only to concentrated loads, there is no distributed load between two consecutive load points.

Under the ideal cable assumptions, each unloaded portion of the cable takes a **straight-line configuration**.

Therefore, a cable carrying only concentrated loads consists of several straight segments connected at the points where the concentrated loads are applied.

Key idea: Concentrated loads produce a cable profile made of **straight segments** between the load points.

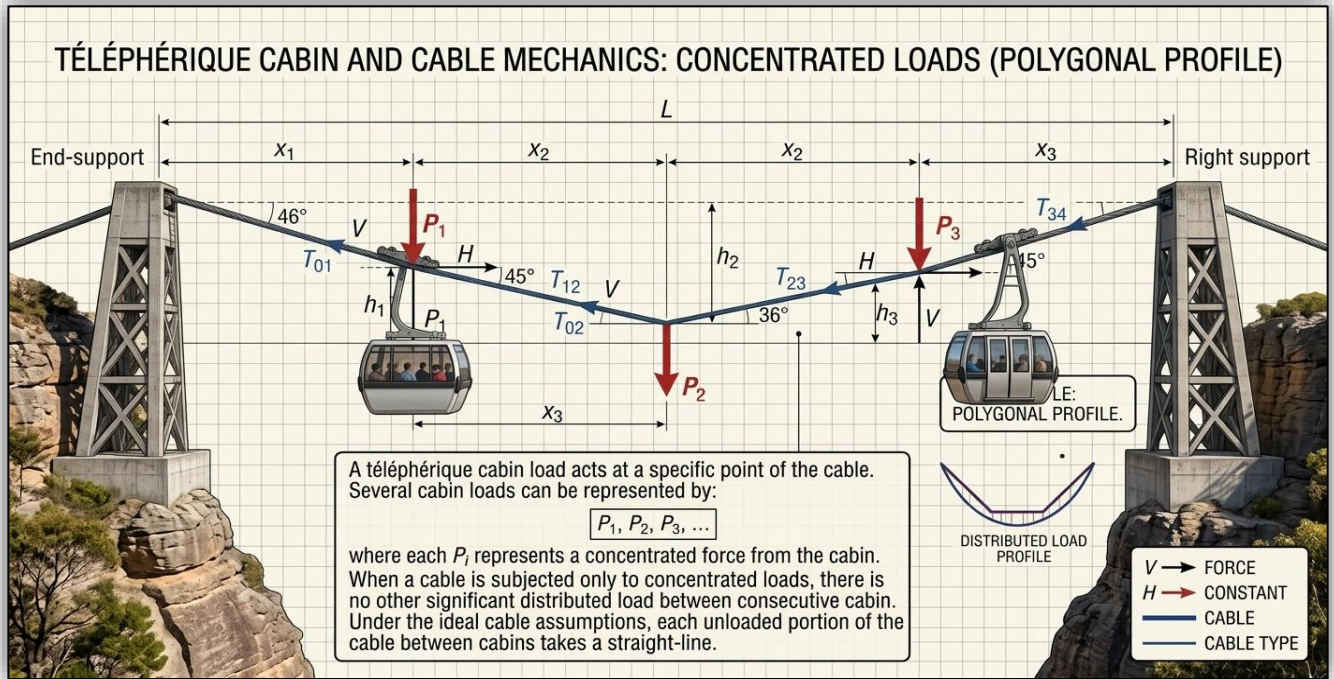


Figure 26— Concentrated loads -cable car (aerial cableway)

A cable may be subjected to several **concentrated loads** ($P_1, P_2, \dots, P_n$). As a first approximation, the self-weight of the cable is assumed to be negligible compared with the applied concentrated loads. The shape of a cable depends on the loading condition: under a uniformly distributed load, the cable takes a **parabolic or catenary shape**, whereas under concentrated loads, it forms a **polygonal line**. If the coordinates of one point of application are known, the **equilibrium equations** can be used to determine the coordinates of the other points and the tensile forces in the successive cable segments. The total cable length can then be calculated as the sum of the lengths of all segments. The **inverse problem**, which consists of determining the positions of the load application points for a prescribed cable length, is more complex and requires the introduction of a **corresponding comparison beam**.

Two methods for analyzing cables subjected to concentrated loads will be presented

1. Direct Method

Consider a cable subjected to a system of concentrated forces as shown in **Figure 27**. The system is **statically indeterminate**.

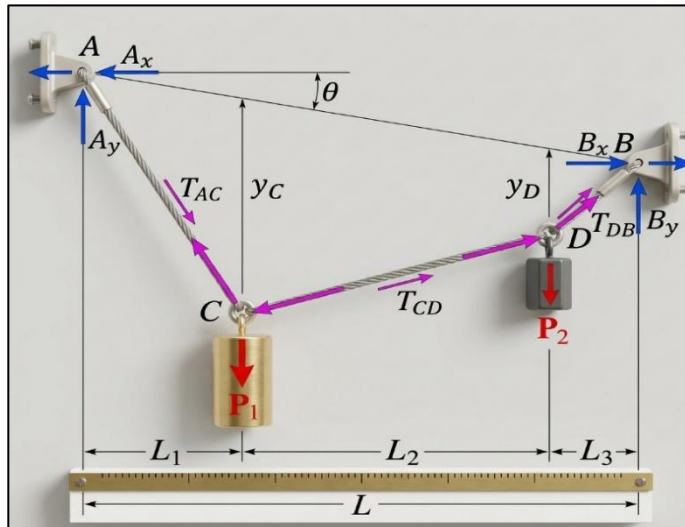


Figure 27- Free-body diagram and geometry of a suspended cable system

For the cable shown in the figure, subjected to two concentrated loads P_1 and P_2 , the system is **statically indeterminate**.

Unknowns

There are **9 unknowns**:

- **Four support reactions:**

$$A_x, A_y, B_x, B_y$$

- **Three cable tensions:**

$$T_{AC}, T_{CD}, T_{DB}$$

- **Two cable sags (vertical coordinates):**

$$y_C, y_D$$

Therefore,

$$\boxed{\text{Number of unknowns} = 4 + 3 + 2 = 9}$$

- **Equilibrium equations**

At each point A , B , C , and D , two equilibrium equations can be written:

$$\sum F_x = 0, \sum F_y = 0$$

Setting the vertical coordinate of support B as $y_B = L \tan \theta$ (where $L = L_1 + L_2 + L_3$):

1. **Node A:**

$$\begin{cases} \sum F_x = 0 \Rightarrow A_x - T_{AC} \cos \theta_1 = 0 \\ \sum F_y = 0 \Rightarrow A_y - T_{AC} \sin \theta_1 = 0 \end{cases}$$

2. **Node C:**

$$\begin{cases} \sum F_x = 0 \Rightarrow T_{AC} \cos \theta_1 - T_{CD} \cos \theta_2 = 0 \\ \sum F_y = 0 \Rightarrow T_{AC} \sin \theta_1 - T_{CD} \sin \theta_2 - P_1 = 0 \end{cases}$$

3. **Node D:**

$$\begin{cases} \sum F_x = 0 \Rightarrow T_{CD} \cos \theta_2 - T_{DB} \cos \theta_3 = 0 \\ \sum F_y = 0 \Rightarrow T_{CD} \sin \theta_2 - T_{DB} \sin \theta_3 - P_2 = 0 \end{cases}$$

4. **Node B:**

$$\begin{cases} \sum F_x = 0 \Rightarrow B_x - T_{DB} \cos \theta_3 = 0 \\ \sum F_y = 0 \Rightarrow B_y - T_{DB} \sin \theta_3 = 0 \end{cases}$$

• **Expressing All Unknowns in Terms of Horizontal Tension H**

From the horizontal equilibrium equations ($\sum F_x = 0$), the **horizontal component of tension is constant** throughout the entire cable:

$$A_x = B_x = T_{AC} \cos \theta_1 = T_{CD} \cos \theta_2 = T_{DB} \cos \theta_3 = H$$

Thus,

$$4 \times 2 = 8 \text{ equilibrium equations}$$

Since there are **9 unknowns** but only **8 independent equilibrium equations**, one additional equation is required.

Geometrical compatibility equation

To make the system **statically determinate**, one additional **geometric equation** must be introduced.

There are two possible alternatives:

• **Using the total cable length**

If the total cable length L_c is known:

$$L_c = \sqrt{L_1^2 + y_C^2} + \sqrt{L_2^2 + (y_D - y_C)^2} + \sqrt{L_3^2 + y_D^2}$$

This equation expresses the total length of the three cable segments:

$$AC + CD + DB = L_c$$

• **Using a prescribed sag**

If the sag at C or D is known, an additional equation is directly available:

$$y_C = \text{known}$$

or

$$y_D = \text{known}$$

Hence, the complete system contains:

$$9 \text{ unknowns} = 8 \text{ equilibrium equations} + 1 \text{ geometric equation}$$

Therefore, the system can be solved.

Exercice 01 :

Determine the tensile forces in the cable shown in the figure 28 and calculate the height h .

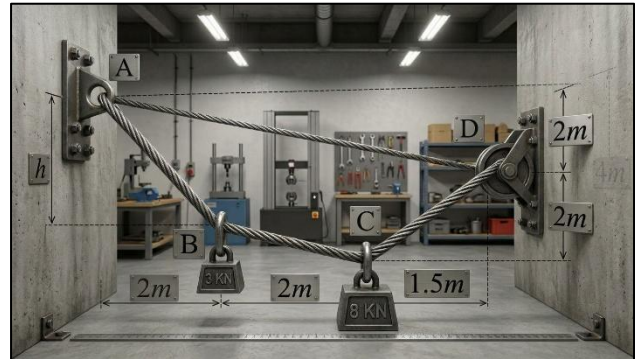
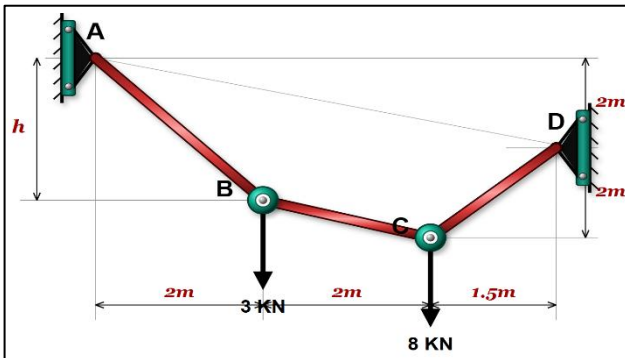


Figure 28 - Suspended cable system under concentrated point loads and realistic simulation of a steel cable in equilibrium

Solution 01 :

1. Geometry and Unknowns

Setting the origin at node $A(0,0)$:

- **Node A:** $(x_A, y_A) = (0, 0)$
- **Node B:** $(x_B, y_B) = (2, -h)$
- **Node C:** $(x_C, y_C) = (4, -4)$ (prescribed sag from diagram: $2\text{ m} + 2\text{ m} = 4\text{ m}$ below level A)
- **Node D:** $(x_D, y_D) = (5.5, -2)$

Applied Loads:

- $P_1 = 3\text{ kN}$ (at Node B)
- $P_2 = 8\text{ kN}$ (at Node C)

2. Slopes of Cable Segments

The tangents of the inclination angles relative to the horizontal are expressed as:

$$\begin{aligned}\tan \theta_1 &= \frac{y_A - y_B}{x_B - x_A} = \frac{0 - (-h)}{2} = \frac{h}{2} \\ \tan \theta_2 &= \frac{y_B - y_C}{x_C - x_B} = \frac{-h - (-4)}{2} = \frac{4 - h}{2} \\ \tan \theta_3 &= \frac{y_D - y_C}{x_D - x_C} = \frac{-2 - (-4)}{1.5} = \frac{2}{1.5} = \frac{4}{3}\end{aligned}$$

3. Nodal Equilibrium Equations Expressed in Terms of H

From horizontal equilibrium ($\sum F_x = 0$), the horizontal component of tension H is constant throughout the entire cable:

$$A_x = D_x = T_{AB} \cos \theta_1 = T_{BC} \cos \theta_2 = T_{CD} \cos \theta_3 = H$$

From vertical equilibrium ($\sum F_y = 0$) at each node:

At Node B:

$$T_{AB} \sin \theta_1 - T_{BC} \sin \theta_2 - P_1 = 0$$

Dividing by H (since $\frac{T \sin \theta}{H} = \tan \theta$):

$$\tan \theta_1 - \tan \theta_2 = \frac{P_1}{H}$$

$$\frac{h}{2} - \frac{4-h}{2} = \frac{3}{H} \Rightarrow \frac{2h-4}{2} = \frac{3}{H} \Rightarrow H(h-2) = 3 \quad \text{--- (Equation 1)}$$

At Node C:

$$T_{CD} \sin \theta_3 + T_{BC} \sin \theta_2 - P_2 = 0$$

Dividing by H :

$$\tan \theta_3 + \tan \theta_2 = \frac{P_2}{H}$$

$$\frac{4}{3} + \frac{4-h}{2} = \frac{8}{H} \Rightarrow \frac{8+3(4-h)}{6} = \frac{8}{H} \Rightarrow H(20-3h) = 48 \quad \text{--- (Equation 2)}$$

4. Solving the System

From Equation 1, $H = \frac{3}{h-2}$. Substituting this into Equation 2:

$$\frac{3}{h-2}(20-3h) = 48$$

$$3(20-3h) = 48(h-2)$$

$$60-9h = 48h-96$$

$$57h = 156 \Rightarrow h = \frac{156}{57} = \frac{52}{19} \approx 2.74 \text{ m}$$

Calculating Horizontal Tension H :

$$H = \frac{3}{\frac{52}{19}-2} = \frac{3}{\frac{14}{19}} = \frac{57}{14} \approx 4.07 \text{ kN}$$

5. Cable Segment Tensions

The tension in any segment i is given by $T_i = \frac{H}{\cos \theta_i} = H\sqrt{1 + \tan^2 \theta_i}$:

1. Segment AB:

$$\tan \theta_1 = \frac{2.7368}{2} = 1.3684$$

$$T_{AB} = 4.0714 \times \sqrt{1 + (1.3684)^2} \approx 6.90 \text{ kN}$$

2. Segment BC:

$$\tan \theta_2 = \frac{4-2.7368}{2} = 0.6316$$

$$T_{BC} = 4.0714 \times \sqrt{1 + (0.6316)^2} \approx 4.82 \text{ kN}$$

3. Segment CD:

$$\tan \theta_3 = \frac{4}{3} \approx 1.3333$$

$$T_{CD} = 4.0714 \times \sqrt{1 + \left(\frac{4}{3}\right)^2} = 4.0714 \times \frac{5}{3} \approx 6.79 \text{ kN}$$

6. Support Reactions

• At Node A:

$$A_x = H = 4.07 \text{ kN}$$

$$A_y = H \tan \theta_1 = 4.0714 \times 1.3684 \approx 5.57 \text{ kN}$$

- At Node D:

$$D_x = H = 4.07 \text{ kN}$$

$$D_y = H \tan \theta_3 = 4.0714 \times \frac{4}{3} \approx 5.43 \text{ kN}$$

Solution 02 :

1. Unknowns and System Equations

- 8 Unknowns: 4 support reactions (X_A, Y_A, X_D, Y_D), 3 cable segment tensions (T_{AB}, T_{BC}, T_{CD}), and sag $h = y_B$.
- 8 Equilibrium Equations: 2 equations ($\sum F_x = 0$ and $\sum F_y = 0$) at each of the 4 nodes (A, B, C, D).

2. Segment CD Geometry and Moment Equilibrium

For segment CD:

- Horizontal span: $\Delta x = 1.5 \text{ m}$
- Vertical span: $\Delta y = 2.0 \text{ m}$
- Length of segment: $L_{CD} = \sqrt{1.5^2 + 2.0^2} = 2.5 \text{ m}$

Trigonometric components:

$$\cos \theta_{CD} = \frac{1.5}{2.5} = 0.6$$

$$\sin \theta_{CD} = \frac{2.0}{2.5} = 0.8$$

Support reactions at node D in terms of tension T_{CD} :

$$X_D = T_{CD} \cos \theta_{CD} = 0.6T_{CD}$$

$$Y_D = T_{CD} \sin \theta_{CD} = 0.8T_{CD}$$

Applying overall moment equilibrium about support A ($\sum M_A = 0$):

$$\sum M_A = X_D \cdot (2) + Y_D \cdot (5.5) - 3 \cdot (2) - 8 \cdot (4) = 0$$

$$(0.6T_{CD}) \cdot 2 + (0.8T_{CD}) \cdot 5.5 - 6 - 32 = 0$$

$$1.2T_{CD} + 4.4T_{CD} = 38$$

$$5.6T_{CD} = 38 \Rightarrow T_{CD} = \frac{38}{5.6} \approx 6.79 \text{ kN}$$

3. Equilibrium of Node C

From the Free Body Diagram of Node C:

- Horizontal axis ($\sum F_x = 0$):

$$T_{CD} \cos \theta_{CD} - T_{BC} \cos \theta_{BC} = 0$$

$$6.79 \times 0.6 - T_{BC} \cos \theta_{BC} = 0 \Rightarrow T_{BC} \cos \theta_{BC} = 4.07 \text{ kN}$$

- Vertical axis ($\sum F_y = 0$):

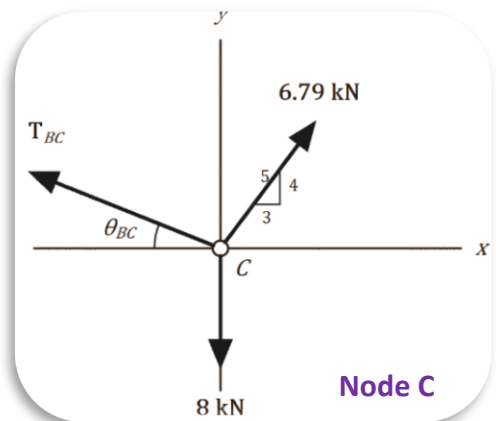
$$T_{CD} \sin \theta_{CD} - 8 + T_{BC} \sin \theta_{BC} = 0$$

$$6.79 \times 0.8 - 8 + T_{BC} \sin \theta_{BC} = 0$$

$$\Rightarrow 5.43 - 8 + T_{BC} \sin \theta_{BC} = 0$$

$$\Rightarrow T_{BC} \sin \theta_{BC} = 2.57 \text{ kN}$$

- Angle θ_{BC} and tension T_{BC} :



$$\tan \theta_{BC} = \frac{2.57}{4.07} \approx 0.6316 \Rightarrow \theta_{BC} = \arctan(0.6316) \approx 32.3^\circ$$

$$T_{BC} = \sqrt{(4.07)^2 + (2.57)^2} \approx 4.82 \text{ kN}$$

4. Equilibrium of Node B

From the Free Body Diagram of Node B:

- Horizontal axis ($\sum F_x = 0$):

$$-T_{BA} \cos \theta_{BA} + T_{BC} \cos \theta_{BC} = 0$$

$$T_{BA} \cos \theta_{BA} = 4.82 \times \cos(32.3^\circ) = 4.07 \text{ kN}$$

- Vertical axis ($\sum F_y = 0$):

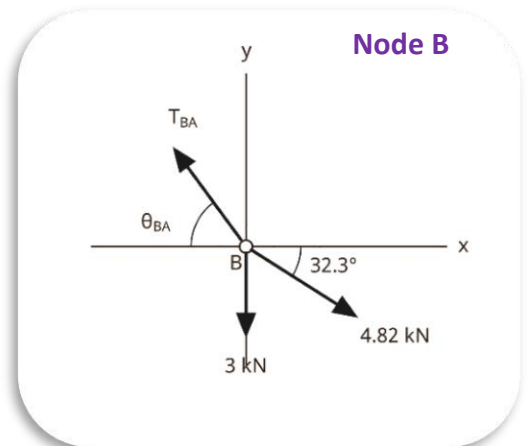
$$T_{BA} \sin \theta_{BA} - T_{BC} \sin \theta_{BC} - 3 = 0$$

$$T_{BA} \sin \theta_{BA} = 4.82 \times \sin(32.3^\circ) + 3 = 2.57 + 3 = 5.57 \text{ kN}$$

- Angle θ_{BA} and tension T_{BA} :

$$\tan \theta_{BA} = \frac{5.57}{4.07} \approx 1.3686 \Rightarrow \theta_{BA} = \arctan(1.3686) \approx 53.8^\circ$$

$$T_{BA} = \sqrt{(4.07)^2 + (5.57)^2} \approx 6.90 \text{ kN}$$



5. Sag Height h Calculation

Using triangle geometry for segment AB:

$$\tan \theta_{BA} = \frac{h}{2 \text{ m}}$$

$$h = 2 \text{ m} \times \tan(53.8^\circ) = 2 \times 1.3686 \approx 2.74 \text{ m}$$

2. Method 02 :Analysis Method – Comparison Beam

Basic Principle: General Cable Theorem

The theorem establishes an equivalence between the behavior of a **cable** and that of a **simply supported virtual beam**:

Fundamental Rule: At any point x along the cable, the product of the horizontal force (**thrust H**) and the **sag/height $y(x)$** is equal to the **bending moment $M_0(x)$** calculated on the equivalent simply supported beam:

$$H \cdot y(x) = M_0(x) \Leftrightarrow y(x) = \frac{M_0(x)}{H}$$

For a cable of negligible weight subjected to vertical loads, the product of the **horizontal component of tension (H)** and the **sag ($y(x)$)** measured at a section (x) is equal to the **bending moment ($M(x)$)** at the same section of a simply supported beam having the same horizontal span and subjected to the same loads as the cable:

La correction LaTeX est :

$$H y(x) = M(x)$$

$$y(x) = M(x)/H$$

This fundamental relationship is known as the **comparison beam principle**.

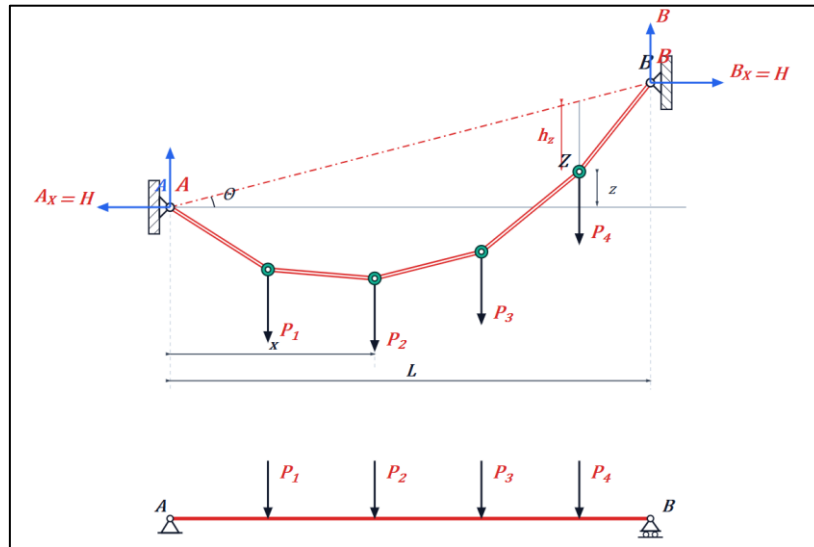
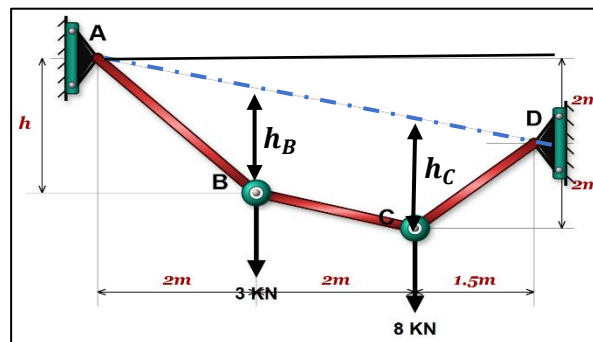


Figure 29-

Example 02

Revisit Example 1 (Figure 28) by applying the general cable theorem to determine H :



The objective is to use this theorem to find the horizontal thrust H and then determine the height h_B at point B.

Step 1: Analyze the equivalent simply supported beam (bottom-right diagram)

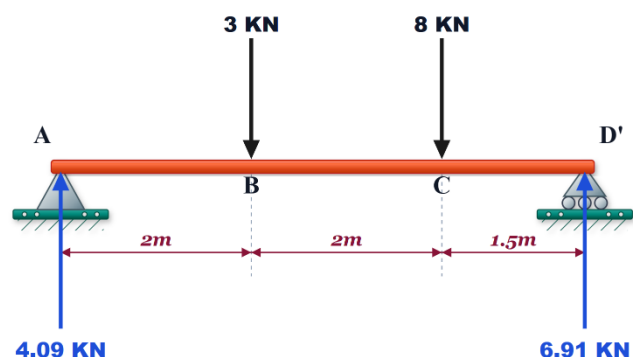
Virtually replace the cable with a straight beam resting on supports A and D with a total span $L = 2 + 2 + 1.5 = 5.5$ m:

- **Applied loads:** 3 kN at B (2 m from A) and 8 kN at C (1.5 m from D).
- **Calculated vertical support reactions:**
 $V_A = 4.09$ kN and $V_D = 6.91$ kN.

Step 2: Calculate the horizontal thrust H

To determine H , point C is used since its height geometry h_C is defined:

1. **Geometric calculation of h_C :**



$$h_C = 2 + 1.5 \times \frac{2}{5.5} = 2.545 \text{ m}$$

2. Calculate bending moment M_C on the virtual beam at point C:

Isolating the right segment (from support D):

$$M_C = V_D \times 1.5 = 6.91 \times 1.5 = 10.365 \text{ kN} \cdot \text{m}$$

3. Deduce horizontal thrust H : According to the theorem, $H \times h_C = M_C$:

$$H = \frac{M_C}{h_C} = \frac{10.365}{2.545} = 4.072 \text{ kN}$$

(The horizontal reaction components at the supports are therefore $A_X = D_X = H = 4.072 \text{ kN}$).

Step 3: Calculate height h_B at point B

Now that H is known, apply the same relation to point B:

1. Calculate bending moment M_B on the virtual beam at point B:

Isolating the left segment (from support A):

$$M_B = V_A \times 2 = 4.09 \times 2 = 8.18 \text{ kN} \cdot \text{m}$$

2. Calculate height h_B :

$$H \times h_B = M_B \Rightarrow h_B = \frac{M_B}{H}$$

$$h_B = \frac{8.18}{4.072} = 2.01 \text{ m}$$

Example 03

Problem Statement: A cable supports two concentrated vertical loads at points B and C (Figure 29).

- Vertical load at B: $P_B = 100 \text{ kN}$
- Vertical load at C: $P_C = 80 \text{ kN}$
- Horizontal distances : $x_{AB} = 2 \text{ m}$, $x_{BC} = 8 \text{ m}$, $x_{CD} = 3 \text{ m}$
- Vertical sag at point C: $y_C = 10 \text{ m}$

Determine:

1. Support reactions at A and D (A_x, A_y, D_x, D_y).
2. Sag at point B (y_B).
3. Tensions in cable segments AB, BC, and CD.
4. Total length of the cable (L)

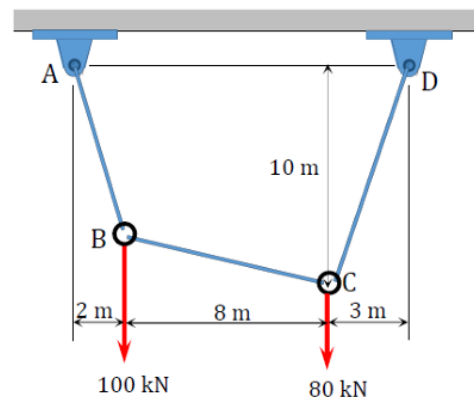


Figure 29 -

Solution

Step 1: Support Reactions (A_x, A_y, D_x, D_y)

1. Total horizontal span:

$$L = x_{AB} + x_{BC} + x_{CD} = 2 + 8 + 3 = 13 \text{ m}$$

2. Vertical Reactions (A_y, D_y):

Taking moments about support A:

$$\sum M_A = 0 \Rightarrow (100 \times 2) + (80 \times 10) - (D_y \times 13) = 0$$

$$200 + 800 - 13D_y = 0 \Rightarrow D_y = \frac{1000}{13} \approx 76.92 \text{ kN}$$

Vertical equilibrium:

$$\sum F_y = 0 \Rightarrow A_y + D_y = 100 + 80 = 180 \text{ kN}$$

$$A_y = 180 - 76.92 = 103.08 \text{ kN}$$

3. Horizontal Reactions (A_x, D_x):

Isolating section CD and taking moments about point C (where $y_C = 10 \text{ m}$):

$$\sum M_C = 0 \Rightarrow (D_y \times 3) - (D_x \times y_C) = 0$$

$$76.923 \times 3 - 10D_x = 0 \Rightarrow D_x = \frac{230.769}{10} \approx 23.08 \text{ kN}$$

Horizontal equilibrium:

$$\sum F_x = 0 \Rightarrow A_x = D_x = H = 23.08 \text{ kN}$$

Step 2: Sag at Point B (y_B)

Isolating section AB and taking moments about point B :

$$\sum M_B = 0 \Rightarrow (A_y \times 2) - (A_x \times y_B) = 0$$

$$103.077 \times 2 - 23.077 \times y_B = 0$$

$$y_B = \frac{206.154}{23.077} \approx 8.93 \text{ m}$$

Step 3: Cable Tensions (T_{AB}, T_{BC}, T_{CD})

1. Tension in segment AB (T_{AB}):

$$T_{AB} = \sqrt{A_x^2 + A_y^2} = \sqrt{23.08^2 + 103.08^2} = \sqrt{532.68 + 10625.49} \approx 105.63 \text{ kN}$$

2. Tension in segment BC (T_{BC}):

- Horizontal force: $H = 23.08 \text{ kN}$
- Vertical shear force: $V_{BC} = A_y - 100 = 103.08 - 100 = 3.08 \text{ kN}$

$$T_{BC} = \sqrt{H^2 + V_{BC}^2} = \sqrt{23.08^2 + 3.08^2} = \sqrt{532.68 + 9.49} \approx 23.28 \text{ kN}$$

3. Tension in segment CD (T_{CD}):

$$T_{CD} = \sqrt{D_x^2 + D_y^2} = \sqrt{23.08^2 + 76.92^2} = \sqrt{532.68 + 5916.69} \approx 80.31 \text{ kN}$$

Step 4: Total Cable Length (L_{total})

Using Pythagorean theorem on each segment:

• Length of segment AB (L_{AB}):

$$L_{AB} = \sqrt{\Delta x_{AB}^2 + y_B^2} = \sqrt{2^2 + 8.93^2} = \sqrt{4 + 79.74} \approx 9.15 \text{ m}$$

- Length of segment BC (L_{BC}):

$$\Delta y_{BC} = y_C - y_B = 10 - 8.93 = 1.07 \text{ m}$$

$$L_{BC} = \sqrt{\Delta x_{BC}^2 + \Delta y_{BC}^2} = \sqrt{8^2 + 1.07^2} = \sqrt{64 + 1.14} \approx 8.07 \text{ m}$$

- Length of segment CD (L_{CD}):

$$L_{CD} = \sqrt{\Delta x_{CD}^2 + \Delta y_{CD}^2} = \sqrt{3^2 + 10^2} = \sqrt{9 + 100} \approx 10.44 \text{ m}$$

- Total Cable Length (L_{total}):

$$L_{\text{total}} = L_{AB} + L_{BC} + L_{CD} = 9.15 + 8.07 + 10.44 = 27.66 \text{ m}$$

II.4. Cable subjected to a uniform horizontally distributed load

Cables provide an effective means of supporting the **dead weight of girders or bridge decks with very long spans**. A **suspension bridge** is a typical example, where the deck is suspended from the main cable by a series of closely spaced, equally distributed hangers.

To analyze this problem, we first determine the shape of a cable subjected to a **uniformly distributed vertical load w_0 per unit horizontal length**, as shown in Fig. 30-a.

The x - and y -axes have their origin at the **lowest point O** of the cable, where the slope is zero. The **free-body diagram** of a small cable segment of length Δs is shown in Fig. 30-b. Since the tensile force in the cable varies continuously in both **magnitude and direction** along its length, the tension is denoted by T . The uniformly distributed load is replaced by its **resultant force $w_0 x$** , which acts at a distance $x/2$ from point O .

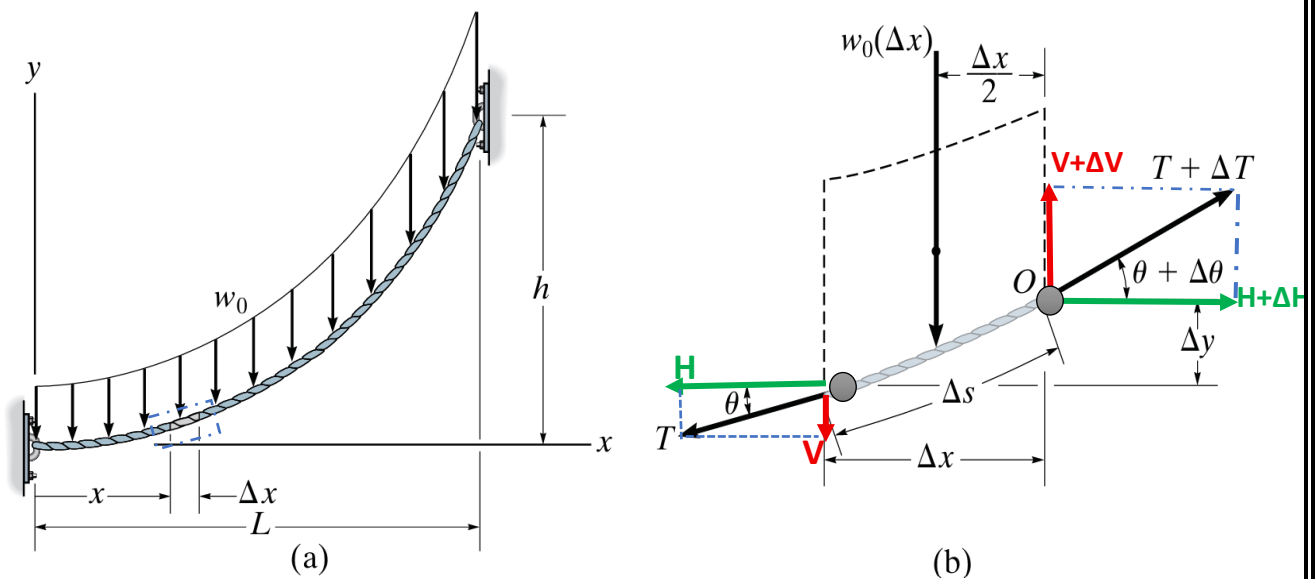


Figure 30- a) Cable subjected to a uniformly distributed vertical load w_0 per unit horizontal length. **b)** Free-body diagram of a small cable segment of length Δs .

Applying the **equations of equilibrium** gives:

1. Horizontal Equilibrium

The horizontal equilibrium of a small cable segment Δs gives:

$$\sum F_x = 0$$

Therefore,

$$-H + (H + dH) = 0$$

which gives:

$$dH = 0$$

Hence, the horizontal component of the cable tension is constant along the cable:

2. Vertical Equilibrium

The vertical equilibrium of a small cable segment ds gives:

$$\sum F_y = 0$$

Thus,

$$-V - w_0 dx + (V + dV) = 0$$

which gives:

$$dV = w_0 dx$$

or

$$\frac{dV}{dx} = w_0(x) \quad (2)$$

From the geometry of the cable:

$$\tan \theta = \frac{dy}{dx} = \frac{V}{H} \quad (3)$$

Since H is constant, differentiating Equation (3) with respect to x gives:

$$\frac{d^2y}{dx^2} = \frac{1}{H} \frac{dV}{dx}$$

Using Equation (2):

$$\frac{d^2y}{dx^2} = \frac{w_0(x)}{H} \quad (4)$$

Therefore, the general equation governing the cable profile is:

$$\frac{d^2y}{dx^2} = \frac{w_0(x)}{H} \quad (5)$$

This equation shows that **the shape of the cable depends directly on the applied distributed load $w_0(x)$**

Solving differential Equation (4), together with the corresponding boundary conditions, gives the cable configuration $(y(x))$, known as the **equilibrium shape** of the cable.

Integrating twice yields:

$$y(x) = \frac{w_0}{2H} x^2 + C_1 x + C_2 \quad (6)$$

For a cable pinned at both ends at the same elevation (0,0) and (L, 0) with maximum sag f at midspan $x = L/2$:

$$y(x) = \frac{4f}{L^2} x(L - x) \quad (7)$$

Where the horizontal tension H is related to the sag by:

$$H = \frac{w_0 L^2}{8f} \quad (8)$$

3. Calculation of Total Cable Tension

The total tension T at any point along the cable is the resultant of its horizontal component H and vertical component V :

$$T^2 = H^2 + V^2$$

Factoring out H^2 :

$$T^2 = H^2 \left(1 + \frac{V^2}{H^2} \right)$$

From the geometry of the cable, the slope is defined as $y' = \frac{dy}{dx} = \tan \theta = \frac{V}{H}$. Substituting $\frac{V}{H} = y'$ gives:

$$T^2 = H^2 (1 + (y')^2)$$

Taking the square root yields the expression for total cable tension $T(x)$:

$$T(x) = H \sqrt{1 + \left(\frac{dy}{dx} \right)^2} = H \sqrt{1 + (y')^2} \quad (9)$$

Physical Remark

- The horizontal component $H = T \cos \theta$ remains **constant** along the entire length of the cable.
- The total tension $T(x)$ directly depends on the local slope y' .
- **The steeper the cable slope** (larger $|y'|$), **the greater the tension T** .
- Consequently, tension T reaches its **maximum at the support/anchor points** (where the slope is steepest) and its **minimum at the lowest point** of the cable (where the slope is zero, $y' = 0 \Rightarrow T = H$).

The horizontal component H is obtained using the boundary condition $y = h$ at $x = L$:

$$H = \frac{w_0 L^2}{2h} \quad \text{---} \quad (10)$$

Finally, substituting equation (10) into equation (4) gives:

$$\frac{d^2 y}{dx^2} = \frac{2h}{L^2} \quad \text{donc } y = h \left(\frac{x}{L} \right)^2 \quad \text{---} \quad (11)$$

4. Maximum Cable Tension

The maximum tension in the cable occurs when the angle θ is maximum, which is at the right support $x = L$. From the force triangle components, we have:

$$T_{\max} = \sqrt{H^2 + (T \sin \theta)^2} \quad \text{---} \quad (12)$$

Since $T \sin \theta = V(x) = w_0 x$, at $x = L$ the vertical reaction is $V_{\max} = w_0 L$. Substituting this gives:

$$T_{\max} = \sqrt{H^2 + (w_0 L)^2} \quad \text{---} \quad (14)$$

5. Arc Length of a Cable

The length s of the cable between two points x_1 and x_2 is given by:

$$s = \int_{x_1}^{x_2} ds$$

From differential geometry:

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + (y')^2} dx$$

Substituting ds into the integral gives the exact total length:

$$s = \int_{x_1}^{x_2} \sqrt{1 + (y')^2} dx$$

This integral can be evaluated either **analytically** or **numerically**, depending on the complexity of the cable profile $y(x)$.

Small-Sag Approximation ($f \ll L$)

When the maximum sag f is small compared to the span L ($f \ll L$), the slope $y' = \frac{dy}{dx}$ is small compared to unity ($(y')^2 \ll 1$). We can simplify the square root using the binomial series expansion $(1 + u)^n \approx 1 + nu$:

$$\sqrt{1 + (y')^2} = (1 + (y')^2)^{1/2} \approx 1 + \frac{1}{2}(y')^2$$

Substituting this approximation into the length integral yields:

$$s \approx \int_{x_1}^{x_2} \left[1 + \frac{1}{2} \left(\frac{dy}{dx}\right)^2 \right] dx = (x_2 - x_1) + \frac{1}{2} \int_{x_1}^{x_2} \left(\frac{dy}{dx}\right)^2 dx$$

6. Application to a Parabolic Cable

For a symmetrical parabolic cable with span L over $x \in [0, L]$ and profile $y(x) = \frac{4f}{L^2} x(L - x)$:

1. Derivative (slope):

$$\frac{dy}{dx} = \frac{4f}{L^2} (L - 2x)$$

2. Integral of $(y')^2$:

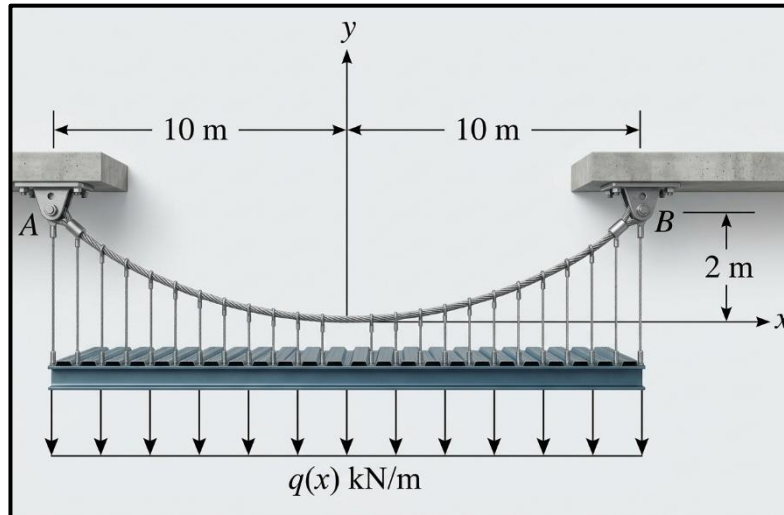
$$\int_0^L \left(\frac{dy}{dx}\right)^2 dx = \frac{16f^2}{L^4} \int_0^L (L - 2x)^2 dx = \frac{16f^2}{3L}$$

3. Approximate Cable Length:

$$s \approx L + \frac{8f^2}{3L}$$

II.4. 1 Special Case: Uniformly Distributed Load

- The most common example of this case is the **main cable of a suspension bridge**. The weight of the bridge is approximately **uniformly distributed horizontally**.
- The **self-weight of the cable** is neglected compared with the weight it supports.
- The load transmitted to the main cable by a large number of **vertical suspenders** can be modeled as a **uniformly distributed load**.



Shape of the Cable

For a **uniformly distributed load** q_0 , the cable takes the shape of a **parabola**.

Starting from Equation (4):

$$q(x) = q_0$$

Since the horizontal component of the cable tension H is constant :

$$y'' = \frac{q_0}{H}$$

Integrating twice gives:

$$y = \frac{q_0}{2H} x^2 + C_1 x + C_2$$

Thus, the equation of the cable is a **parabola**.

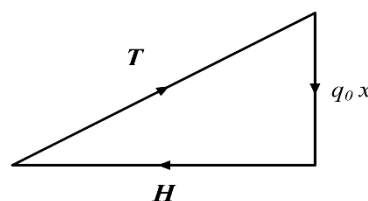
The constants C_1 and C_2 are determined from the **support conditions**.

Here, x is the horizontal coordinate measured from the **lowest point of the cable**.

Tension in the Cable

At a section located at a horizontal distance x from the lowest point, the tension is:

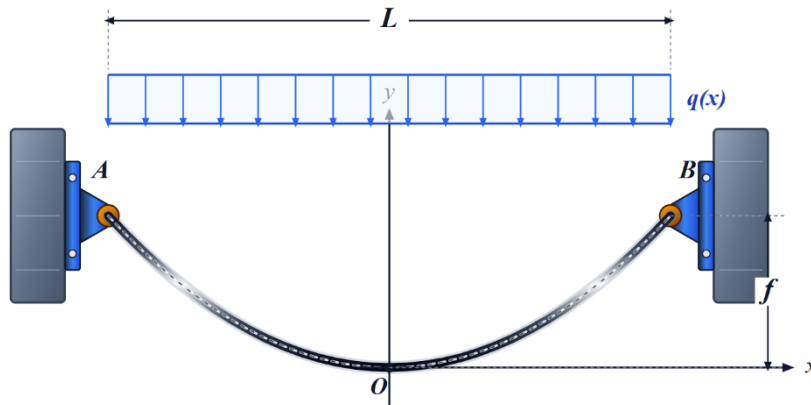
$$T = \sqrt{H^2 + (q_0 x)^2}$$



where:

- H = constant horizontal component of the cable tension;
- q_0 = uniformly distributed load per unit horizontal length;
- x = horizontal distance from the lowest point;
- T = total tension in the cable.

Application 1: Horizontal Cable under a Uniformly Distributed Load



For $x = 0$, we have:

$$y = y' = 0$$

Therefore,

$$C_1 = C_2 = 0$$

and the cable equation becomes:

$$y = \frac{q_0}{2H} x^2$$

At the supports A and B, the sag is f , and the span is L . Thus, for $x = \frac{L}{2}$:

$$f = \frac{q_0}{2H} \left(\frac{L}{2}\right)^2$$

Hence,

$$H = \frac{q_0 L^2}{8f}$$

Since

$$y = \frac{q_0}{2H} x^2$$

we obtain:

$$y = \frac{4fx^2}{L^2}$$

Tension in the Cable

The tension at a section located at x is:

$$T = \sqrt{H^2 + q_0^2 x^2}$$

Using

$$H = \frac{q_0 L^2}{8f}$$

gives:

$$T = \frac{q_0 L^2}{8f} \sqrt{1 + \left(\frac{8fx}{L^2}\right)^2}$$

The maximum tension occurs at the supports *A* and *B*, where:

$$x = \frac{L}{2}$$

Therefore:

$$T_{\max} = \sqrt{H^2 + \left(\frac{q_0 L}{2}\right)^2}$$

Introducing the parameter

$$k = \frac{4f}{L}$$

we obtain:

$$T_{\max} = H \sqrt{1 + k^2}$$

Or, equivalently,

$$T_{\max} = H \sqrt{1 + \left(\frac{4f}{L}\right)^2}$$

Cable Length

The length of the cable over the horizontal interval from 0 to *x* is:

$$s = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Since

$$y = \frac{4fx^2}{L^2}$$

we have:

$$\frac{dy}{dx} = \frac{8fx}{L^2}$$

Therefore,

$$s = \int_0^x \sqrt{1 + \left(\frac{8fx}{L^2}\right)^2} dx$$

Using the binomial expansion and retaining the first three terms:

$$\sqrt{1+u} \approx 1 + \frac{1}{2}u - \frac{1}{8}u^2$$

we obtain:

$$s \approx \int_0^x \left[1 + \frac{1}{2}\left(\frac{8fx}{L^2}\right)^2 - \frac{1}{8}\left(\frac{8fx}{L^2}\right)^4 \right] dx$$

For the **total cable length**, by symmetry:

$$L_{\text{cable}} = 2 \int_0^{L/2} \left[1 + \frac{1}{2}\left(\frac{8fx}{L^2}\right)^2 - \frac{1}{8}\left(\frac{8fx}{L^2}\right)^4 \right] dx$$

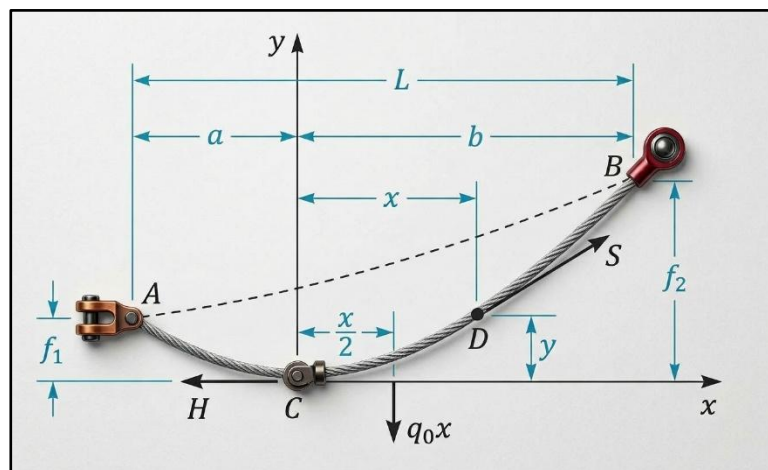
After integration:

$$L_{\text{cable}} \approx L + \frac{8f^2}{3L} - \frac{32f^4}{5L^3}$$

Or, in dimensionless form:

$$L_{\text{cable}} = L \left\{ 1 + \frac{8}{3}\left(\frac{f}{L}\right)^2 - \frac{32}{5}\left(\frac{f}{L}\right)^4 \right\}$$

Application 2: Inclined Cable and Uniformly Distributed Load



For a cable pinned at its two ends at **different levels**, we obtain:

For $x = 0$, we have $y = y' = 0$, hence

$$C_1 = C_2 = 0 \quad \text{and} \quad y = \frac{q_0}{2H} x^2$$

At A, $x = a$ and $y = f_1$, hence $f_1 = \frac{q_0 a^2}{2H}$

At B, $x = b$ and $y = f_2$, hence $f_2 = \frac{q_0 b^2}{2H}$

The equilibrium of segment CD implies $T = \sqrt{H^2 + (q_0 x)^2}$

In particular, the tensions at A and B are:

$$T_A = \sqrt{H^2 + (q_0 a)^2} \quad \text{and} \quad T_B = \sqrt{H^2 + (q_0 b)^2}$$

1. Differential Length of the Cable

For an infinitesimal cable segment:

$$ds = \sqrt{dx^2 + dy^2}$$

Dividing by dx :

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Hence:

$$ds = \sqrt{1 + \left(\frac{q_0 x}{H}\right)^2} dx$$

2. Length of the Cable Segment CB

Let b be the horizontal distance between the lowest point C and support B .

The cable length between C and B is therefore :

$$s_{CB} = \int_C^B ds$$

or:

$$s_{CB} = \int_0^b \sqrt{1 + \left(\frac{q_0 x}{H}\right)^2} dx$$

To evaluate this integral, introduce the substitution :

$$u = \frac{q_0 x}{H}$$

Thus:

$$x = \frac{H}{q_0} u$$

and:

$$dx = \frac{H}{q_0} du$$

The limits become:

$$\begin{aligned} x = 0 &\Rightarrow u = 0 \\ x = b &\Rightarrow u = \frac{q_0 b}{H} \end{aligned}$$

Therefore:

$$s_{CB} = \frac{H}{q_0} \int_0^{q_0 b/H} \sqrt{1+u^2} du$$

3. Evaluation of the Integral

The required standard integral is:

$$\int \sqrt{1+u^2} du = \frac{1}{2} \left[u\sqrt{1+u^2} + \operatorname{asinh}(u) \right]$$

where:

$$\operatorname{asinh}(u) = \ln \left(u + \sqrt{1+u^2} \right)$$

Thus:

$$s_{CB} = \frac{H}{2q_0} \left[u\sqrt{1+u^2} + \operatorname{asinh}(u) \right]_0^{q_0 b/H}$$

Substituting the upper limit:

$$s_{CB} = \frac{H}{2q_0} \left[\frac{q_0 b}{H} \sqrt{1 + \left(\frac{q_0 b}{H} \right)^2} + \operatorname{asinh} \left(\frac{q_0 b}{H} \right) \right]$$

Therefore:

$$s_{CB} = \frac{b}{2} \sqrt{1 + \left(\frac{q_0 b}{H} \right)^2} + \frac{H}{2q_0} \operatorname{asinh} \left(\frac{q_0 b}{H} \right)$$

4. Length of the Cable Segment CA

Similarly, let a be the horizontal distance between the lowest point C and support A .

Following exactly the same procedure:

$$s_{CA} = \int_0^a \sqrt{1 + \left(\frac{q_0 x}{H} \right)^2} dx$$

Therefore:

$$s_{CA} = \frac{a}{2} \sqrt{1 + \left(\frac{q_0 a}{H} \right)^2} + \frac{H}{2q_0} \operatorname{asinh} \left(\frac{q_0 a}{H} \right)$$

5. Total Length of the Cable AB

The total cable length is the sum of the two portions:

$$L_{\text{cable}} = s_{CA} + s_{CB}$$

Hence:

$$L_{\text{cable}} = \frac{a}{2} \sqrt{1 + \left(\frac{q_0 a}{H}\right)^2} + \frac{H}{2q_0} \operatorname{asinh}\left(\frac{q_0 a}{H}\right) + \frac{b}{2} \sqrt{1 + \left(\frac{q_0 b}{H}\right)^2} + \frac{H}{2q_0} \operatorname{asinh}\left(\frac{q_0 b}{H}\right)$$

This is the **exact expression** for the length of a parabolic cable subjected to a uniformly distributed load over its horizontal projection.

6. Alternative Logarithmic Form

Since

$$\operatorname{asinh}(u) = \ln(u + \sqrt{1 + u^2}),$$

The segment CB can also be written as:

$$s_{CB} = \frac{b}{2} \sqrt{1 + \left(\frac{q_0 b}{H}\right)^2} + \frac{H}{2q_0} \ln \left[\frac{q_0 b}{H} + \sqrt{1 + \left(\frac{q_0 b}{H}\right)^2} \right]$$

And similarly for CA .

7. Special case: symmetric cable

If the cable is symmetric about its lowest point C :

$$a = b = \frac{L}{2}$$

where L is the horizontal span.

Then:

$$L_{\text{cable}} = 2s_{CB}$$

and therefore:

$$L_{\text{cable}} = L \sqrt{1 + \left(\frac{q_0 L}{2H}\right)^2} + \frac{H}{q_0} \operatorname{asinh}\left(\frac{q_0 L}{2H}\right)$$

This is the **exact cable length** for the symmetric parabolic cable.